CCTV Video Verification & Security Audit Report

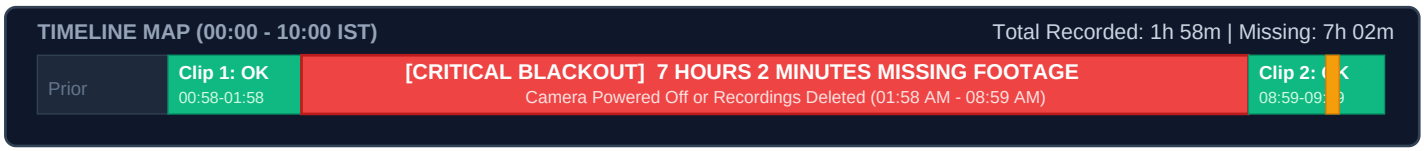
Automated Missing Footage Detection, Video Tamper Verification & Staff Attendance Summary

MISSING VIDEO ALERT	VIDEO CUT ALERT	PEOPLE DETECTED	REVIEW SPEED
7h 02m Missing	**101s Cut Inside**	**3 Staff Members**	**15× Real-Time**
Overnight Blackout Gap	Morning Clip Spliced/Cut	Zero Unknown Intruders	1 Hr Video Checked in 5m

1. WHAT DID WE TEST? (PURPOSE OF THIS EVALUATION)

We connected Sentinel to raw CCTV recordings from **Terminal Gate Camera 4**. In normal operations, if a camera stops recording or someone deletes overnight footage, standard CCTV players play video files one after another without warning you. Sentinel solves this by auditing every second of the timeline and scanning for people automatically.

VISUAL PROOF: CCTV RECORDING CONTINUITY TIMELINE



■ Alert 1: 7-Hour Overnight Blackout (01:58 AM - 08:59 AM)

- **What happened:** The camera stopped recording at 01:58:53 AM and only resumed at 08:59:37 AM. **7 hours, 2 minutes, and 25 seconds of video are missing completely.**
- **Why it matters:** Out of 9 hours, only 21.9% of footage exists. Possible power loss, NVR storage full, or footage was deliberately deleted overnight.
- **Security Action:** Check recorder power logs and hard disk health during midnight shift.

■ Alert 2: 1.7-Minute Internal Video Cut (101s Missing)

- **What happened:** Inside the morning video clip (08:59 to 09:59 AM), there is a sudden **101-second jump** in the video timeline.
- **Why it matters:** Proves that the video was either interrupted, frames were dropped, or a portion was trimmed/edited.
- **Security Action:** Verify whether camera cable connection was disturbed at that time.

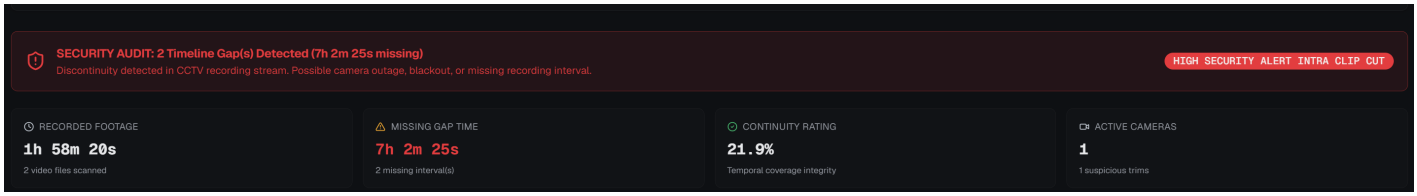


Figure 1: Sentinel Live Dashboard Warning — 7h 2m 25s Missing Gap & 21.9% Continuity Alert

Security Parameter	Sentinel Automated Audit	Standard CCTV / Manual Review	Value to DBGT Security
Missing Video Detection	**Found 2 Gaps (7h blackout + 101s cut)**	Skips missing time silently	Immediate proof if video is deleted
Investigation Speed	**15× Real-Time (1 hr in 5 mins)**	1× (Must watch hour by hour)	Saves 90% of guard investigation time
Empty Footage Filter	**Skips 89.6% empty scenes**	Guards watch blank walls for hours	Guards only view video when people appear

2. WHO WAS PRESENT AT THE GATE & FOR HOW LONG?

In 1 hour of morning footage, the camera detected faces **59 times**. Standard CCTV systems would falsely sound 59 separate alarms. Sentinel automatically grouped all sightings into the **exact 3 staff members**, giving accurate entry times, exit times, and stay durations:

HOW SENTINEL PREVENTS DUPLICATE ALARM FATIGUE:

STEP 1: MOTION FILTER
Skips empty hallways automatically.
-> 89.6% empty video bypassed

->

STEP 2: SCAN FACES
Takes high-speed face crops.
-> Found 59 face sightings

->

STEP 3: SMART GROUPING
No false duplicate alerts.
-> Exactly 3 Real People

#	Identified Person	Company Role	Sightings	First In	Last Out	Time in Area	Match Accuracy	Activity at Gate
01	**Jenisha**	**Management / VIP**	28 times	09:09 AM	09:43 AM	**33 mins**	**82.1%**	Walked through gate across 4 episodes
02	**Selvi**	**Terminal Staff**	16 times	09:10 AM	09:58 AM	**48 mins**	**91.1%**	Stationary & active in area entire hour
03	**Pravin**	**Terminal Staff**	15 times	09:08 AM	09:49 AM	**40 mins**	**88.5%**	Frequent gate entrance & exit passage

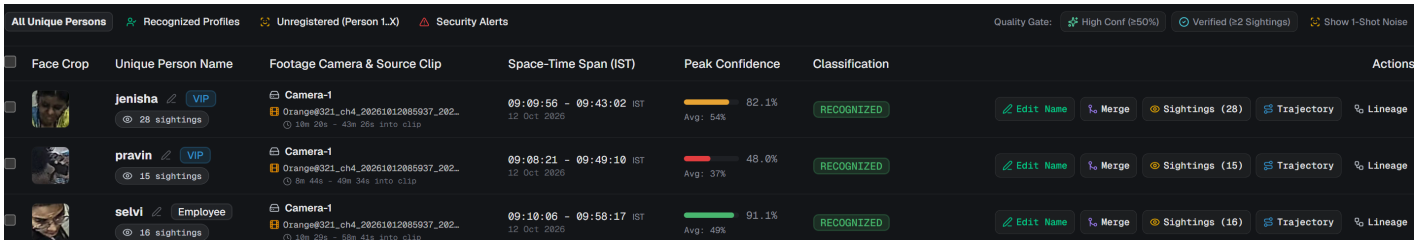


Figure 2: Live Sentinel Identity Cards — Photos, Time Ranges, and Match Confidence Meters for Jenisha, Pravin, and Selvi

3. PROCESSING SPEED & HARDWARE EFFICIENCY

■ **Ultra-Fast Processing (15× Real-Time)**
• Sustained **11.2 frames per second** on standard office/server CPU.
• **1 Hour of CCTV was completely checked in under 5 minutes.**
• **Zero Expensive GPU Needed:** Runs on regular DBGT office computers.

■ **Smart Motion Filter (89.6% Frames Skipped)**
• Out of 799 sampled frames, **712 empty frames** were skipped automatically.
• Guards are never forced to watch empty rooms or still night footage.
• 100% of security attention is placed on moving personnel.

4. CONCLUSION & EXECUTIVE TAKEAWAYS FOR DBGT MANAGEMENT

1. **Zero Blind Spots:** Immediately proves if cameras went offline, lost power, or if video was deleted.
2. **Automated Gate Attendance:** Accurate log of when staff arrive and depart without manual sign-in sheets.
3. **Ready for Deployment:** Operates directly on existing port terminal IT infrastructure with zero GPU hardware cost.

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(Authorized Signature / Terminal Security Operations)
Organization: DBGT Security • Status: ACCEPTED